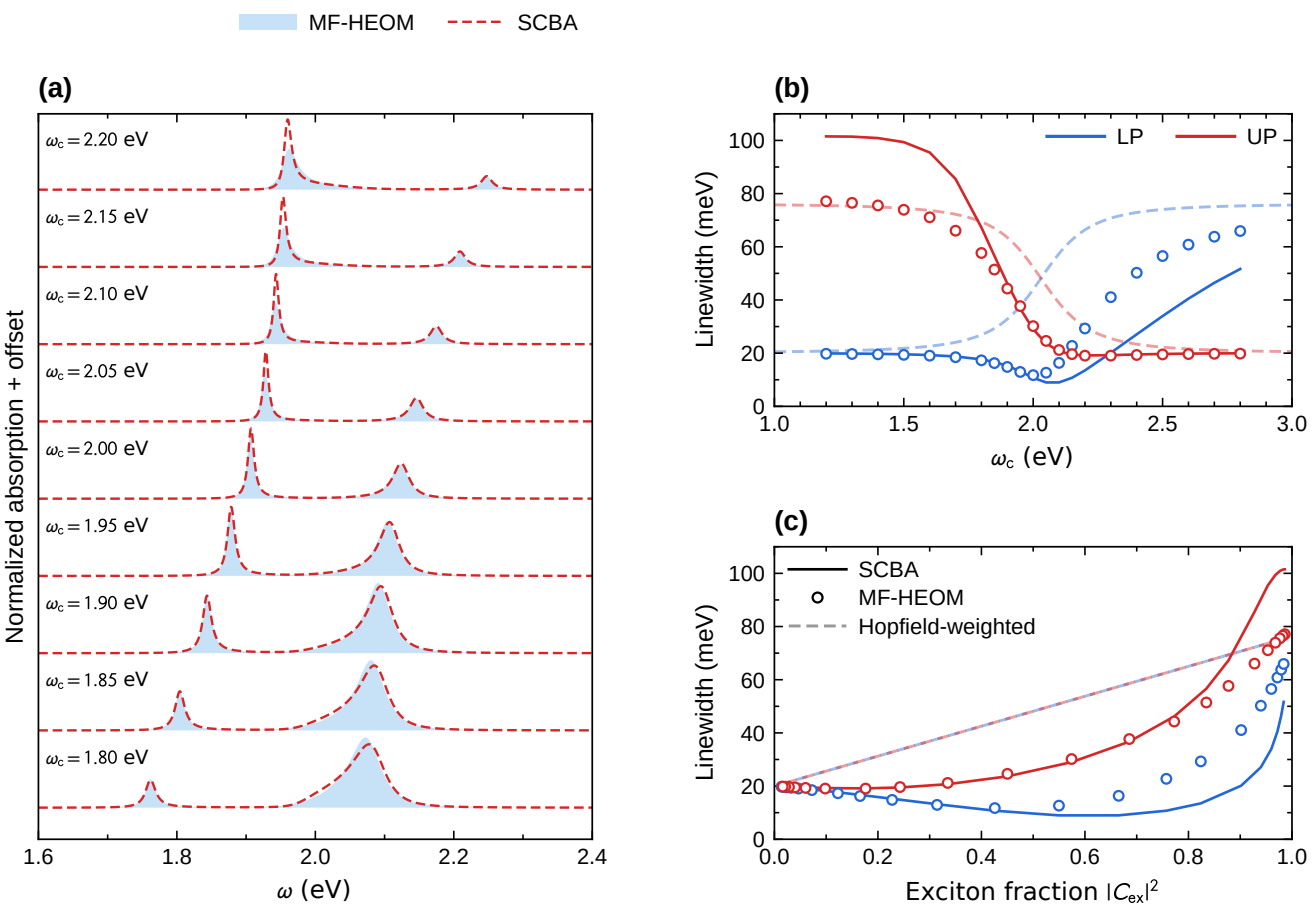

MF-HEOM
SCBA
(a)
$\omega_c = 2.20$ eV
$\omega_c = 2.15$ eV
$\omega_c = 2.10$ eV
$\omega_c = 2.05$ eV
$\omega_c = 2.00$ eV
$\omega_c = 1.95$ eV
$\omega_c = 1.90$ eV
$\omega_c = 1.85$ eV
$\omega_c = 1.80$ eV
Normalized absorption + offset
1.6
1.8
2.0
2.2
2.4
ω (eV)
(b)
LP
UP
Linewidth (meV)
0
20
40
60
80
100
1.0
1.5
2.0
2.5
3.0
ω_c (eV)
(c)
SCBA
MF-HEOM
Hopfield-weighted
Linewidth (meV)
0
20
40
60
80
100
0.0
0.2
0.4
0.6
0.8
1.0
Exciton fraction $|C_{ex}|^2$